Run report: sessions C and D (stride-renew targets), 2026-10-14 17:28Z

Study: `research/ecbench_rootfind_speedup_20261014` (Conductor T-1). **Protocol:** `AMENDMENT_1.md`, committed before these runs. **Binary:** `ecbench` SHA-256 `9a45707c6a59a426…`. **Host:** 4 vCPU Intel Xeon 2.10 GHz, shared cloud container; `--cpus none --allow-busy`, isolation **L0** on every record. **Runner:** `run_amendment1.sh`, which first re-audited sessions A and B with this binary (ok, 768 and 144 records).

session	spec	curve	targets × rounds	records
C `ECBS1h97bbbf34584f`	`claim-koblitz-rootfind-stride-renew-n13.json`	`icv1-f2m13-t181-515ee569`, r = 2003	32 × 3 (+1 warm-up)	896 / 896 verified
D `ECBS1hb8d9ed954ea3`	`claim-koblitz-rootfind-stride-renew-n19.json`	`icv1-f2m19-t797-b6cf2467`, r = 130873	16 × 3 (+1 warm-up)	256 / 256 verified

Audits (`verify --replay-all`): C ok, 896 records; D ok, 256 records. No run failed or timed out.

Measured S (GAE / √r), mean over measured runs, 95 % interval as `ecbench table` recorded

arm	C (n = 13)	D (n = 19)	priced
rho-strong	14.289 [14.148, 14.437]	3.131 [3.038, 3.224]	lower bound
rho-frob	1.927 [1.810, 2.122]	—	lower bound
d5-walk / d6-walk	23.508 [22.545, 24.504]	17.552 [16.129, 18.902]	C full; D lower bound
d5-stride / d6-stride	13.048 [12.525, 13.594]	23.426 [20.989, 25.808]	C full; D lower bound
d5-renew / d6-renew	12.483 [12.141, 12.823]	14.657 [13.217, 16.134]	C full; D lower bound
d4-renew	11.005 [10.297, 11.762]	—	full
rho-strong-aa	14.289 [14.148, 14.437]	—	lower bound

Paired comparisons (`ecbench compare`, candidate ÷ baseline, cluster interval)

- C: d5-renew ÷ d5-walk = 0.5310 [0.506, 0.557], 96 pairs on 32 workloads.
- C: d4-renew ÷ rho-strong = 0.7702 [0.720, 0.824] (diagnostic only: rho-strong has unpriced work).
- C: d5-renew ÷ d5-stride = 0.9567 [0.910, 1.004].
- C: d5-renew ÷ rho-strong = 0.8736 [0.847, 0.900] (diagnostic only).
- C: rho-strong-aa ÷ rho-strong = 1.0000 [1.000, 1.000].
- D: d6-renew ÷ d6-walk = 0.8351 [0.741, 0.943], 48 pairs on 16 workloads (diagnostic only: both arms leave lookups and row operations unpriced on this uncalibrated curve).
- D: d6-renew ÷ d6-stride = 0.6257 [0.545, 0.713] (diagnostic only).
- D: d6-renew ÷ rho-strong = 4.6813 [4.191, 5.202]; d6-walk ÷ rho-strong = 5.6058 [5.122, 6.082].

Wall medians are L0 and descriptive only.

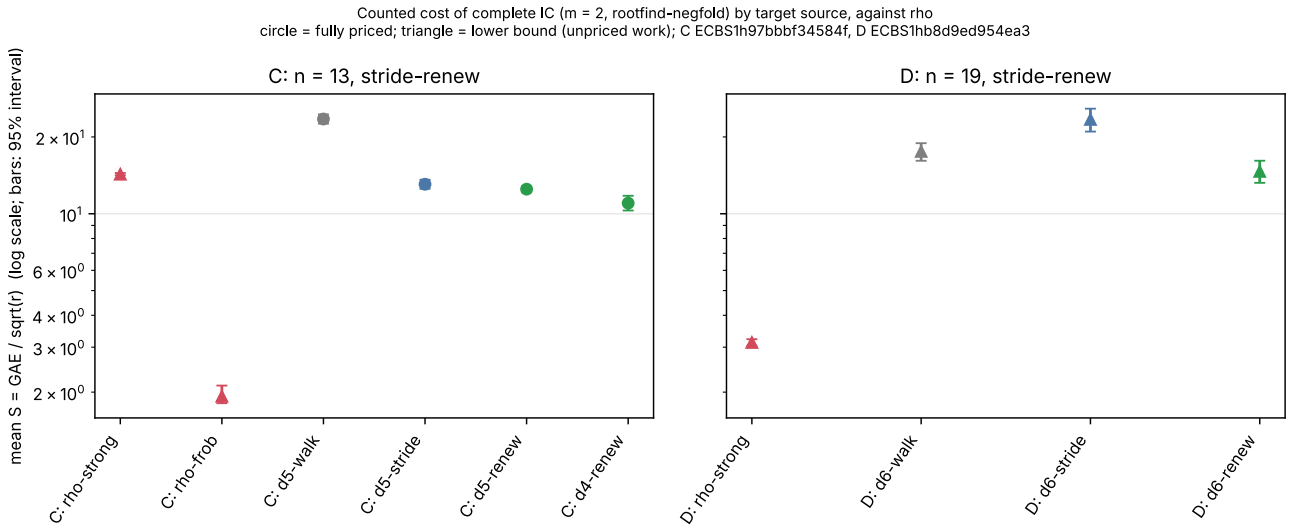


Figure 1: Mean S with the recorded 95 % interval per arm, sessions C (`ECBS1h97bbbf34584f`) and D (`ECBS1hb8d9ed954ea3`); circles fully priced, triangles lower bounds.

Raw records: `sessions/claim-renew-n13/records.jsonl`, `sessions/claim-renew-n19/records.jsonl`; tables and comparisons under `tables/`.